

RESEARCH ARTICLE

Dynamic Forecast-Optimal Smoothness for Penalized Trend Estimation

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Abstract

A penalized trend estimator has a continuum of admissible smoothness values, but selecting one for extrapolation is a different task from recovering a historical signal. Moreover, chronological forecast-loss surfaces may contain several local minima that evolve through time. We construct a time-indexed representation of these minima by matching nearby candidates across forecast origins and recording each persistent branch through its smoothness values, initial validation loss, and subsequent refitted forecast loss. The resulting branch state separates two decisions: which branch to use and how to map its history into the smoothness selected for the next forecast. This provides a common representation for recent means and medians, loss- and recency-weighted rules, and extrapolations of the smoothness trajectory. Controlled simulations support horizon-matched tuning of the pooled benchmark. A small confirmation panel favors a recency-based branch rule, but a larger financial panel and changing-roughness simulations do not establish general superiority over pooled forecast cross-validation. A separate simulation illustrates how alternative branch functionals differ in prediction and boundary behavior. The contribution is a transparent, reproducible framework for constructing and evaluating multiple smoothness decisions from tracked forecast-loss minima, rather than a universally preferred selection rule.

KEYWORDS

forecast evaluation; penalized least squares; smoothing parameter; trend estimation; time-series cross-validation; rolling-origin forecasting

1 | INTRODUCTION

Smoothing parameters are often treated as technical constants, although in forecasting they determine a substantive choice: how much recent movement is treated as signal and how much is suppressed before extrapolation. Finite-difference penalized least squares (PLS) makes this trade-off especially transparent because the penalty can be translated into an interpretable normalized smoothness scale.

Guerrero (Guerrero, 2007, 2008) develops this controlled-smoothness view. The forecasting problem begins once smoothness is no longer specified *ex ante*. A natural first approach is to choose the value that minimizes historical chronological forecast loss. That pooled selector remains an important baseline in this

paper, but it ignores a feature that becomes visible when the loss surface is inspected repeatedly through time: the same surface can contain several local minima, and those minima can persist, drift, appear, or disappear across neighboring forecast origins.

The paper therefore studies forecast-optimal smoothness dynamically. At each chronological origin t , let

$$\mathcal{M}_t = \{S_{1,t}, \dots, S_{K,t}\}$$

be the local minima of the origin-specific forecast-loss surface. Nearby minima on adjacent surfaces are connected into data-driven branches. A branch is represented by its smoothness trajectory together with the Validation-1 loss at which the local minimum is identified and the subsequent Validation-2 forecast loss obtained after refitting through Validation 1.

This produces a branch state

$$V_j = [S_{j,t}, \ell_{j,t}^{(1)}, \ell_{j,t}^{(2)}]_t.$$

The forecasting decision then has two stages. First, a historical rule ψ selects a persistent branch. Second, a rule ϕ maps the selected branch history into the current smoothness:

$$\hat{j}_T = \psi(V_1, \dots, V_J), \quad \hat{S}_T = \phi(V_{\hat{j}_T}).$$

The simplest dynamic rule uses the newest tracked minimum. Other admissible maps can use a recent mean or median, Validation-2-weighted smoothness, recency weights, or forecasts of the smoothness trajectory itself. They operate on the same branch history; the purpose of presenting several examples is to make the decision space explicit, not to define a universal best rule.

The older pooled criterion remains useful as a benchmark. For historical origins $T_1, \dots, T_M$,

$$F_{d,L,h}^{\text{pool}}(S) = \frac{1}{M} \sum_{m=1}^M F_{T_m}(S),$$

with

$$\hat{S}_{d,L,h}^{\text{pool}} \in \arg \min_S F_{d,L,h}^{\text{pool}}(S).$$

It answers which single smoothness value has the lowest average historical forecast loss. The dynamic framework retains the temporal organization of multiple local optima and makes it available to a family of smoothness decisions. Whether any particular decision improves future forecasts remains an empirical question.

The forecasting literature provides strong precedents for choosing tuning parameters by predictive performance. Forecast-error minimization is standard in exponential smoothing (Taylor, 2004); Hart (Hart, 1994) studies predictive smoothing selection in dependent data; and more recent forecasting work continues to connect tuning and out-of-sample loss (Zafar, Kellard, & Vinogradov, 2022; Staněk, 2023; Wolff & Echterling, 2024; Franjic & Schweikert, 2025; Xu, Aschakulporn, & Zhang, 2025). The present contribution is narrower: it focuses on the normalized smoothness of finite-difference penalized trends and the temporal evolution of multiple minima of their forecast-loss surfaces.

The distinction from numerical optimization is also explicit. The companion numerical paper studies how to recover all relevant local minima of each surface and how to correspond them across time. This paper studies what those tracked branches mean for forecasting

and how they should be converted into the current smoothness.

Finally, validation never supplies the trend fit used in the final forecast. Once $\hat{S}_T$ is chosen, all temporary validation fits are discarded, the trend is refit on the newest L observations available at T , and only then is the untouched future block forecast.

The empirical record is informative but mixed. Horizon-matched pooled selection improves forecast accuracy in the controlled baseline simulation. A frozen branch-based recency rule performs favorably in a small confirmation panel but not in a broader financial panel; simulations under changing latent roughness also fail to establish a general improvement over pooled forecast-CV. The principal contribution is thus the explicit separation between recovering and tracking multiple forecast-loss minima, selecting a branch, and defining a current smoothness decision from that branch. The remainder of the paper develops these objects, reports the frozen empirical results, and illustrates several distinct decision maps under common information restrictions.

2 | RELATED WORK AND POSITIONING

2.1 | Controlled smoothness in finite-difference PLS

The direct methodological foundation is the controlled-smoothness formulation of finite-difference PLS. Guerrero (Guerrero, 2007) derives a statistical representation of the penalized trend estimator and introduces a trace-based smoothness index linking the penalty parameter to an interpretable amount of smoothness. Guerrero (Guerrero, 2008) develops this into a user-controlled trend-estimation procedure in which a desired smoothness level is chosen first and the corresponding penalty is recovered afterward.

Cortés-Toto et al. (Cortés-Toto, Guerrero, & Reyes, 2017) study what smoothness is produced when the PLS penalty is selected by CV, GCV, AICc, and BIC. Their results make an important distinction for the present paper: a rule for selecting λ and the amount of smoothness ultimately induced by that rule are not the same object.

The present paper reverses the controlled-smoothness direction:

$$\text{user chooses } S \longrightarrow \lambda(S)$$

becomes

$$\text{chronological forecast loss} \longrightarrow S^*.$$

The percentage of smoothness remains the interpretable object, but it is selected endogenously for the forecast task.

2.2 | Predictive selection of smoothing and tuning parameters

Forecast-based tuning is established in other smoother families. Hart (Hart, 1994) considers kernel smoothing with serially correlated errors and proposes time-series cross-validation (TSCV), selecting the bandwidth through one-step-ahead predictive performance. This is an important conceptual precedent because it distinguishes good trend recovery from good prediction under dependence.

The object considered here is different. Hart selects a kernel bandwidth and explicitly models serial dependence in the errors. We select the smoothness of a finite-difference PLS trend, retain the model's native continuation rule, and allow an arbitrary h -step future block to define the tuning target. The connection to Hart is therefore the predictive principle rather than estimator equivalence.

Forecast-error tuning is also standard in exponential smoothing. Taylor (Taylor, 2004) describes data-driven smoothing-parameter estimation through ex post one-step-ahead forecast errors and develops an adaptive smooth-transition formulation. More recent *Journal of Forecasting* papers apply the same predictive principle to other tuning problems. Wolff and Echterling (Wolff & Echterling, 2024) choose regularization and machine-learning tuning parameters using validation forecast errors. Franjic and Schweikert (Franjic & Schweikert, 2025) propose a cross-validation strategy that connects predictor preselection directly to nowcast performance. Xu et al. (Xu et al., 2025) compare bandwidth-selection schemes in a time-varying-parameter HAR model and show that bandwidth choice affects forecast performance.

These papers establish that forecast-oriented tuning is a legitimate methodological target. The narrower gap addressed here is the selection of a controlled smoothness percentage inside the finite-difference PLS family.

2.3 | Trend filtering for forecasting

Zafar et al. (Zafar et al., 2022) explicitly evaluate trend-filtering methods by the short-term forecasts they enable. Their study reinforces the idea that a trend representation should be judged partly by predictive consequences rather than only by visual smoothness or in-sample fit. Our objective is complementary: rather than introduce a new trend representation, we keep finite-difference PLS fixed and optimize the amount of smoothness within that family.

2.4 | Chronological forecast evaluation

The proposed criterion uses pseudo-out-of-sample loss as a tuning objective. Tashman (Tashman, 2000) reviews out-of-sample forecast evaluation schemes, while Bergmeir and Benítez (Bergmeir & Benítez, 2012) discuss cross-validation for time-series predictors. Staněk (Staněk, 2023) formalizes out-of-sample loss estimation and distinguishes rolling and fixed evaluation schemes. His formulation also makes explicit that the relevant forecast loss depends on the forecast horizon, model memory, and updating design.

This is closely aligned with $F_{d,L,h}(S)$. Smoothness is selected for a specified horizon and update protocol, and the information set available at each origin is part of the estimator's definition.

3 | FINITE-DIFFERENCE PENALIZED TREND ESTIMATION

3.1 | Penalized least squares

At forecast origin T , let

$$x_T = (y_{T-L+1}, \dots, y_T)^\top \in \mathbb{R}^L$$

denote the estimation window. For difference order $d < L$, let

$$D_d \in \mathbb{R}^{(L-d) \times L}, \quad Q = D_d^\top D_d.$$

The baseline estimator solves

$$\hat{\tau}_T(\lambda) = \arg \min_{\tau \in \mathbb{R}^L} \{ \|x_T - \tau\|_2^2 + \lambda \|D_d \tau\|_2^2 \}, \quad \lambda \geq 0, \quad (1)$$

with solution

$$\hat{\tau}_T(\lambda) = H_\lambda x_T, \quad H_\lambda = (I + \lambda Q)^{-1}. \quad (2)$$

The paper uses the zero-drift, identity-weight formulation in (1) to isolate the smoothness-selection question. Guerrero's broader statistical formulation allows nonzero mean differences and more general covariance weighting (Guerrero, 2007); these are natural extensions but are not silently imposed in the baseline criterion.

3.2 | Spectral representation

Because

$$v^\top Q v = \|D_d v\|_2^2 \geq 0,$$

Q is symmetric positive semidefinite. Under the standard difference operator,

$$\text{rank}(Q) = L - d,$$

so Q has d zero eigenvalues and $L - d$ positive eigenvalues. Write

$$Q = U \text{diag} \left(\underbrace{0, \dots, 0}_d, \delta_1, \dots, \delta_{L-d} \right) U^\top, \quad \delta_j > 0.$$

Then

$$H_\lambda = U \text{diag} \left(\underbrace{1, \dots, 1}_d, \frac{1}{1 + \lambda \delta_1}, \dots, \frac{1}{1 + \lambda \delta_{L-d}} \right) U^\top. \quad (3)$$

Thus the d null-space directions are reproduced without shrinkage, while each penalizable direction is attenuated by

$$\alpha_j(\lambda) = \frac{1}{1 + \lambda \delta_j}.$$

For $d = 2$, the null space consists of discrete linear sequences, so increasing λ moves the fitted path toward a linear trend.

3.3 | Normalized smoothness

Guerrero's trace-based construction motivates measuring smoothing through the effective action of H_λ . We use

$$S(\lambda) = 1 - \frac{1}{L-d} \sum_{j=1}^{L-d} \frac{1}{1 + \lambda \delta_j}. \quad (4)$$

The normalization excludes the d directions that cannot be penalized, so

$$S(0) = 0, \quad \lim_{\lambda \rightarrow \infty} S(\lambda) = 1.$$

For a fixed linear smoother,

$$\text{edf}(\lambda) = \text{tr}(H_\lambda).$$

Combining (3) and (4),

$$\boxed{\text{edf}(\lambda) = L - (L - d)S(\lambda)}. \quad (5)$$

Hence S is a normalized complement of effective degrees of freedom.

4 | FORECAST-OPTIMAL SMOOTHNESS

4.1 | Continuation of the fitted trend

A trend estimate does not become a forecast until a continuation rule is specified. In the baseline model, the fitted finite-difference structure is continued by imposing zero future d th differences. This yields a fixed linear operator

$$G_{d,h} \in \mathbb{R}^{h \times L}$$

such that

$$\hat{z}_T(\lambda) = G_{d,h} H_\lambda x_T. \quad (6)$$

Equivalently, using backward differences at the end of the fitted trend,

$$\hat{\tau}_{T+k|T} = \sum_{j=0}^{d-1} \binom{k+j-1}{j} \nabla^j \hat{\tau}_T, \quad k = 1, \dots, h. \quad (7)$$

For $d = 1$ the continuation is constant; for $d = 2$ it is linear; and for $d = 3$ it is quadratic.

4.2 | Origin-specific forecast-loss surfaces

Let

$$z_T = (y_{T+1}, \dots, y_{T+h})^\top$$

be the realized future block at origin T . For candidate λ ,

$$f_T(\lambda) = \frac{1}{h} \|z_T - G_{d,h} H_\lambda x_T\|_2^2. \quad (8)$$

Because $S(\lambda)$ is one-to-one, we write the same surface in normalized smoothness as

$$F_T(S) = f_T(\lambda(S)), \quad S \in [0, 1].$$

At a chronological sequence of historical forecast origins $T_1, \dots, T_M$, each surface can have one or several local minima. Denote the local-minimum set at origin T_m by

$$\mathcal{M}_m = \{S_{1,m}, \dots, S_{K_m,m}\}. \quad (9)$$

This sequence of local-minimum sets is the central dynamic object of the paper.

4.3 | Pooled forecast-CV as a baseline

A natural static baseline averages forecast losses across historical origins:

$$F_{d,L,h}^{\text{pool}}(S) = \frac{1}{M} \sum_{m=1}^M F_{T_m}(S). \quad (10)$$

Its minimizer is

$$\hat{S}_{d,L,h}^{\text{pool}} \in \arg \min_{S \in [0,1]} F_{d,L,h}^{\text{pool}}(S). \quad (11)$$

This selector asks which single smoothness value has the lowest average historical forecast loss. It remains an important benchmark, but it discards the temporal organization of the local minima themselves.

4.4 | Tracking local minima through time

To retain that organization, we connect nearby local minima across adjacent forecast origins. A current minimum $S_{j,m}$ may continue a previous branch when

$$|S_{j,m} - S_{j,m-1}| \leq \varepsilon, \quad (12)$$

subject to one-to-one matching at each origin.

The radius ε does not define a fixed exogenous region such as “smoothness near 0.9.” Instead, branch locations are determined by the data through the recovered local minima; ε only controls whether minima on neighboring surfaces are close enough to be treated as the same temporal branch.

For a tracked branch j , let $\ell_{j,m}^{(1)}$ denote its Validation-1 forecast loss at the origin where the local minimum is identified. The trend is then refit through Validation 1 using the same $(d, L, S_{j,m})$, and the immediately following block is scored to obtain $\ell_{j,m}^{(2)}$, the Validation-2 forecast loss.

We collect the branch history as

$$V_j = \begin{pmatrix} S_{j,t_1} & \ell_{j,t_1}^{(1)} & \ell_{j,t_1}^{(2)} \\ S_{j,t_2} & \ell_{j,t_2}^{(1)} & \ell_{j,t_2}^{(2)} \\ \vdots & \vdots & \vdots \end{pmatrix}. \quad (13)$$

4.5 | Branch selection and current smoothness

The forecasting decision is separated into two steps. First, a historical branch-selection rule

$$\hat{j}_T = \psi(V_1, \dots, V_J) \quad (14)$$

chooses among persistent branches using information available before the outer test. A simple baseline prefers branches with maximal support and then minimizes mean historical Validation-2 loss.

Second, a branch-to-smoothness rule

$$\hat{S}_T = \phi(V_{\hat{j}_T}) \quad (15)$$

converts the selected branch history into the smoothness used at the current forecast origin.

The simplest rule is the newest tracked minimum,

$$\phi_{\text{last}}(V_j) = S_{j,T}.$$

Alternative stabilizing rules use the recent mean or median,

$$\phi_{\text{mean},K}(V_j) = \frac{1}{K} \sum_{r=0}^{K-1} S_{j,T-r},$$

or

$$\phi_{\text{med},K}(V_j) = \text{median}\{S_{j,T-K+1}, \dots, S_{j,T}\}.$$

An alternative weighted mean uses recency alone. Let $r = 0$ denote the newest tracked smoothness value, with older values indexed by increasing r . Then

$$\phi_{\text{recency}}(V_j) = \frac{\sum_{r=0}^R \rho^r S_{j,T-r}}{\sum_{r=0}^R \rho^r}, \quad 0 < \rho < 1.$$

The newest value receives weight one and older values are exponentially discounted. For interpretability we parameterize ρ by a half-life H ,

$$\rho = 2^{-1/H},$$

so the weight assigned to a value H tracked origins old is one half of the newest weight.

A loss-weighted rule gives more influence to historically predictive values:

$$\phi_{V2}(V_j) = \frac{\sum_t w_{j,t} S_{j,t}}{\sum_t w_{j,t}}, \quad w_{j,t} = \frac{1}{\ell_{j,t}^{(2)} + \delta},$$

with $\delta > 0$ used only for numerical stabilization. Recency may be incorporated through

$$w_{j,t} = \frac{\rho^{T-t}}{\ell_{j,t}^{(2)} + \delta}, \quad 0 < \rho \leq 1.$$

A distinct class of maps treats the branch smoothness values as a series to be extrapolated. For a branch with values $S_{j,t_1}, \dots, S_{j,t_m}$, fit the auxiliary trend

$$(\hat{a}, \hat{b}) \in \arg \min_{a,b} \sum_{u=1}^m \omega_u (S_{j,t_u} - a - bu)^2.$$

The resulting branch-based smoothness decision is

$$\phi_{\text{trend}}(V_j) = \Pi_{[0,1]}(\hat{a} + \hat{b}(m+1)),$$

where $\Pi_{[0,1]}$ truncates to the admissible normalized-smoothness interval. Uniform weights, recent-window weights, or exponentially increasing recency weights lead to different examples. Alternatively, one can extrapolate the branch increments:

$$\phi_{\Delta,\rho}(V_j) = \Pi_{[0,1]} \left[S_{j,t_m} + \frac{\sum_{r=0}^{m-2} \rho^r (S_{j,t_{m-r}} - S_{j,t_{m-r-1}})}{\sum_{r=0}^{m-2} \rho^r} \right].$$

These auxiliary models forecast the next *smoothness value*, not the underlying observations or an old fitted trend. The final observation forecast still follows the fresh-refit procedure below.

The examples above are not intended to define a tournament for one universally best smoothness rule. Their role is to show that the tracked branch state V_j supports a family of admissible decision maps

$$\phi : V_j \mapsto \hat{S}_T.$$

Different applications may emphasize persistence, recency, historical forecast performance, robustness, or forward extrapolation of the smoothness trajectory. Empirical comparisons describe the behavior of these choices under controlled conditions; they do not make one functional part of the definition of the framework.

4.6 | Mandatory final refit

Neither the pooled selector nor the dynamic branch rules retain an old fitted trend. Once the current smoothness $\hat{S}_T$ has been selected, all intermediate validation fits are discarded and the trend is refit using the most recent information available at the outer origin:

$$\hat{\tau}_T = H_{\lambda(\hat{S}_T)} y_{T-L+1:T}. \quad (16)$$

The operational forecast is then

$$\hat{y}_{T+1:T+h|T} = G_{d,h} \hat{\tau}_T.$$

Thus the proposed dynamic method can average historical losses or smoothness values when a declared decision rule requires it, but it never averages historical fitted trends.

4.7 | Relationship to conventional cross-validation

The construction remains distinct from ordinary leave-one-out CV and GCV. The relevant losses are chronological extrapolation losses, and the dynamic extension additionally studies persistence of local minima across successive forecast-loss surfaces.

Hart's TSCV (Hart, 1994) remains an important predictive-smoothing precedent. The present problem differs in estimator, normalized smoothness coordinate, multi-step continuation, multimodal objective geometry, and the explicit temporal tracking of local smoothness minima.

5 | BASIC PROPERTIES

Proposition 1 (Monotone smoothness reparameterization). *For positive penalized eigenvalues $\delta_1, \dots, \delta_{L-d}$,*

$$S'(\lambda) = \frac{1}{L-d} \sum_{j=1}^{L-d} \frac{\delta_j}{(1 + \lambda \delta_j)^2} > 0,$$

and

$$S''(\lambda) = -\frac{2}{L-d} \sum_{j=1}^{L-d} \frac{\delta_j^2}{(1 + \lambda \delta_j)^3} < 0.$$

After adjoining $\lambda = \infty$, S maps $[0, \infty]$ continuously and strictly increasingly onto $[0, 1]$.

Proof. Differentiate (4) term by term. Positivity of each δ_j gives the signs; the endpoint values follow from $(1 + \lambda \delta_j)^{-1}$. $\square$

The bounded scale has a common interpretation across fixed (d, L) configurations while preserving the ordering of penalty strength.

Proposition 2 (Effective degrees of freedom). *For fixed λ ,*

$$\text{tr}(H_\lambda) = L - (L-d)S(\lambda).$$

Proof. From (3),

$$\text{tr}(H_\lambda) = d + \sum_{j=1}^{L-d} (1 + \lambda \delta_j)^{-1}.$$

Substitution of (4) yields the result. $\square$

Proposition 3 (Equivalent optimization in penalty and smoothness space). *Let $F(S) = f(\lambda(S))$. Then the global minimizers in $\lambda \in [0, \infty]$ and $S \in [0, 1]$ correspond under the bijection in Proposition 1. At interior points,*

$$F'(S) = \frac{f'(\lambda)}{S'(\lambda)},$$

so interior stationary points correspond exactly.

Proof. A strictly increasing bijection preserves the minimizing set. The derivative identity follows from the chain rule. $\square$

Proposition 4 (Endpoint interpretation). *At $S = 0$, $H_\lambda = I$ and the historical trend reproduces the observations. At $S = 1$, interpreted as $\lambda \rightarrow \infty$, H_λ converges to the orthogonal projection onto $\ker(D_d)$.*

For $d = 2$, the endpoints are therefore the unsmoothed historical path and its least-squares linear projection. Endpoint comparison is part of the scientific selection problem, not merely a numerical detail.

5.1 | Existence and possible nonuniqueness

For finite $S < 1$, $H_{\lambda(S)}$ is continuous. The limiting matrix at $S = 1$ exists by (3). Thus the forecast loss has a continuous extension to the compact domain $[0, 1]$ and at least one global minimizer exists.

Uniqueness is not assumed. Distinct smoothing regimes can produce similar validation loss, and the criterion may contain several local minima. This is statistically meaningful because competing minima can imply different effective degrees of freedom and terminal slopes. Efficient recovery of all numerical candidates is delegated to the companion numerical-methods paper; the definition in (11) does not depend on a particular optimizer.

5.2 | Forecast optimality versus recovery optimality

If a latent trend τ is known in simulation, a recovery-oriented oracle might choose

$$S_{\text{rec}} \in \arg \min_S \|\hat{\tau}_S - \tau\|^2.$$

The proposed criterion instead chooses $S_{d,L,h}^*$ from future predictive error. These targets need not coincide. A smoother can reconstruct the historical latent path accurately but extrapolate poorly, or sacrifice historical

recovery in exchange for more stable terminal derivatives.

This distinction is a central empirical question of the paper, not a claim that forecast-optimal smoothness is universally preferable under every loss.

6 | EVALUATION DESIGN

The experiments address three distinct questions. First, can historical forecast loss choose a useful smoothness level for a specified horizon? Second, how do tracked-minimum decisions behave when compared with a pooled forecast-loss optimum? Third, what changes when several functionals $\phi(V_j)$ act on the same historically identified branch? We do not predefine a universal winning functional; the third question concerns the expressiveness and limitations of the branch representation.

6.1 | Chronological information and re-fitting

For a forecast origin T , candidate smoothness values are computed from surfaces and branch histories observed before the untouched future block. A completed branch row contains the minimum located using Validation 1, then a Validation-2 score computed after the trend is refit through Validation 1. These scores are available for historical branch rows only. The current final-Validation-1 minimum has no realized Validation-2 score and is not assigned one when loss-weighting past observations.

For every evaluated rule the final trend is re-estimated on the latest L observations available at T . The temporary fits used to find minima or score the validation blocks never enter the operational forecast as historical fitted-trend averages. When forecast origins advance, the information set and the branch histories advance chronologically.

6.2 | Pooled forecast tuning

The pooled benchmark averages historical forecast errors across origins for fixed (d, L, h) and minimizes the resulting objective in normalized smoothness. Checkpoint 03 fixes $d = 2$, $L = 60$, horizons 1, 3, 6, 12, five latent trend mechanisms, and iid Gaussian, AR(1), and Student- t disturbances. It compares horizon-matched forecast-CV with one-step forecast-CV, ordinary CV,

GCV, and AICc, as well as non-feasible simulation-only latent oracles. The simulation reuses innovations across different trend mechanisms within a given seed; uncertainty calculations must therefore respect the seed as the resampling unit.

6.3 | Tracked-minimum evaluation

The branch experiments recover local minima at historical rolling origins, connect nearby candidates with a one-to-one correspondence under a fixed continuation radius, and summarize each branch by persistence and the subsequent Validation-2 loss. The baseline historical selector ψ prefers persistent, predictive branches. Different maps ϕ act on the selected branch while sharing the same numerical candidate recovery, branch identities, and outer test blocks.

Checkpoint 04 used a four-series development and reserved-confirmation design. The recency-weighted rule with three-origin half-life was frozen before evaluating the 16 held-out confirmation blocks. Checkpoint 05 applied it without retuning to 64 other financial series, each with four outer blocks. Its broader results motivated Checkpoint 06, a *post-hoc* analysis of native continuation stability using earlier blocks from the same series. The order restriction in that diagnostic is explanatory; it is not a new external confirmation.

6.4 | Controlled dynamic roughness and the rule family

Checkpoint 07 fixes $d = 2$ and studies 1,200 simulated scenarios with constant, gradually changing, or abruptly switching latent slope-innovation variance, using 100 seeds and two observation-noise scales. It provides a controlled comparison between the pooled selector, the newest local minimum, and recency averaging along a tracked branch.

Checkpoint 08 uses a disjoint set of 100 seeds and the same family of roughness mechanisms. It fixes the underlying branch selection while evaluating several maps ϕ , including linear trends fitted to recent smoothness values, exponentially weighted linear trends, and weighted increments. A raw extrapolated smoothness outside $[0, 1]$ is projected onto that interval before the fresh final fit. The frequency of projection is reported alongside forecast-error ratios, since it is a distinctive behavior of extrapolative functionals.

6.5 | Metrics and interpretation

The financial panels use *level* RMSE; the controlled roughness simulations use *log*-scale RMSE. For comparisons performed on a common outer test block, the summary is the geometric mean of the paired RMSE ratios. Ratios below one favor the numerator's rule, but these ratios must not be compared across incompatible series panels or scales as though they were one pooled experiment.

Uncertainty summaries from the four-series confirmation and 64-series financial panels are descriptive, especially because asset returns share shocks and the confirmation sample has only four series. Results from different checkpoints cannot be treated as repeated independent confirmations of a single rule. The separate numerical-methods paper analyzes the accuracy of minimum recovery and matching; the present paper addresses how their recorded histories may be used as forecasting decisions.

7 | EMPIRICAL EVIDENCE AND EXAMPLES

The empirical program separates predictive tuning of the penalized trend from the more general question of what can be done with a tracked history of forecast-loss minima. All reported results below are taken from the frozen checkpoint outputs in the accompanying repository. Their evaluation designs and loss scales differ, so we do not pool their performance ratios into a single estimate of forecasting accuracy.

7.1 | Selecting smoothness for the forecast horizon

The pooled-selector simulation (Checkpoint 03) comprises 3,000 synthetic scenarios, formed from five latent trends, three noise mechanisms, two noise scales, and 100 random seeds. Each scenario has six chronological forecast origins evaluated at horizons $h \in \{1, 3, 6, 12\}$, giving 72,000 outer-origin-horizon evaluations. The estimator fixes $d = 2$ and $L = 60$.

Table 1 reports the geometric root-mean-square forecast error (RMSFE) ratio of horizon-matched pooled forecast-CV to two feasible selectors. At $h = 1$, horizon-matched and one-step forecast-CV coincide by construction. At longer horizons the matched selector performs better in aggregate, particularly under serially correlated disturbances. These results substantiate the

TABLE 1 Frozen pooled-selector simulation (Checkpoint 03). Geometric RMSFE ratios below one favor horizon-matched forecast-CV.

h	vs. one-step forecast-CV	vs. GCV
1	1.000	0.912
3	0.947	0.816
6	0.861	0.688
12	0.762	0.554

distinction between selecting smoothness for historical trend recovery and selecting it for extrapolation; they do not yet involve temporal branch tracking.

7.2 | Evidence on the tracked-minimum decision

Checkpoint 04 evaluated branch-based smoothness selection on a small heterogeneous panel (GDPC1, SPY, AAPL, and BTC-USD), with the recency-weighted rule ϕ_{recency} fixed from development data before the last four non-overlapping test blocks per series were opened. Over the resulting 16 confirmation decisions, its geometric level-RMSE ratio was 0.692 relative to pooled forecast-CV and 0.814 relative to the newest tracked minimum. It beat pooled CV in 13 of 16 blocks. These are results for this particular frozen four-series test, not a universal advantage of recency weighting.

The much broader external Checkpoint 05 panel comprised 64 previously unused financial series (20 exchange-traded funds, 36 individual stocks, and eight cryptocurrencies) and four test blocks per series. In these 256 decisions, the same frozen recency rule had a geometric *level*-RMSE ratio of 1.642 to pooled forecast-CV, with a descriptive series-cluster bootstrap interval of [1.168, 2.703]. Against using the newest tracked minimum, however, its geometric ratio was 0.525. Thus smoothing the branch trajectory substantially stabilized the newest-minimum decision in this panel, while the pooled selector remained superior overall. The financial series are dependent through common market shocks; the cluster interval is a descriptive sensitivity calculation rather than an independent-series sampling guarantee.

7.3 | The role of continuation order

A post-hoc diagnostic of Checkpoint 05 associated extreme forecast errors with higher-order polynomial

continuation, especially the cubic continuation induced by $d = 4$. Checkpoint 06 examined that mechanism on earlier historical blocks that did not overlap the external test blocks. It compared successively restricted allowable difference orders, holding the remaining branch-based protocol fixed.

The geometric level-RMSE ratios of recency weighting to pooled CV were 1.353 for $d \in \{1, 2, 3, 4\}$, 1.200 for $\{1, 2, 3\}$, 1.073 for $\{1, 2\}$, and 1.045 for $d = 2$ alone. Over the same order sets, the recency/newest-minimum ratios moved from 0.691 to 0.994. The large apparent stabilization benefit is therefore concentrated where native higher-order continuation amplifies relatively small changes in selected smoothness. Since the order restriction was motivated by Checkpoint 05 outcomes, this diagnostic is explanatory, not a separately confirmed revised forecasting method.

7.4 | Controlled changes in latent roughness

Checkpoint 07 fixed $d = 2$, $L = 120$, and $h = 20$ to separate changes in smoothness from high-order continuation instability. It generated 1,200 scenarios with stationary, gradually varying, and abruptly changing latent slope-innovation scales, and evaluated 9,600 chronological outer forecast decisions.

The frozen recency rule yielded geometric *log*-RMSE ratios to pooled forecast-CV of 1.028 under changing roughness and 1.056 under stationary roughness (1.037 across all mechanisms). Its ratio to the newest tracked minimum was 0.945 overall. Hence, even under deliberately changing latent roughness, recency averaging did not establish a general gain relative to pooled forecast-CV. The simulation instead isolates a more modest, recurrent property: it can damp variation in the newest-minimum rule.

7.5 | A family of smoothness decisions from the same branch

Checkpoint 08 uses a fresh set of 100 simulation seeds with the same controlled roughness mechanisms (1,200 scenarios, 9,600 outer decisions). For each decision, the selected branch history is held fixed while its final smoothness is computed by multiple functionals $\phi(V_j)$. Examples include recency averages, least-squares extrapolations fitted to the latest three, five, or ten branch values, exponentially weighted linear extrapolations, and extrapolations of historical branch increments.

TABLE 2 Illustrative branch-to-smoothness functionals (Checkpoint 08). The ratio is geometric observed-log-RMSE relative to pooled forecast-CV; clipping refers to raw predictions outside $[0, 1]$.

Functional	RMSE ratio	Clipped (%)
Recency mean (HL 3)	1.041	0.0
Weighted linear (HL 5)	1.063	10.8
Linear extrapolation ($K = 10$)	1.079	18.4
Weighted increments (HL 3)	1.110	22.3

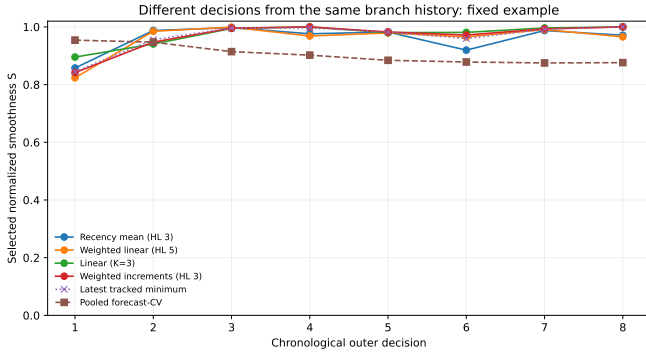


FIGURE 1 Selected normalized smoothness across eight chronological outer decisions under one fixed simulated roughness-switch scenario (seed 100, noise standard deviation 0.01). The rules act on the same recovered branch histories; the example was not selected for favorable forecast performance.

Table 2 gives four illustrative functionals. They differ both in observed forecasting behavior and in how frequently a raw extrapolated smoothness lies outside the admissible interval $[0, 1]$. Clipping is unnecessary for the recency mean, since a convex average of admissible smoothness values is itself admissible, but is needed for some forecast-based maps. All four examples, as well as the other extrapolative rules examined in this checkpoint, had aggregate geometric ratios above one relative to pooled forecast-CV. They are not ranked to identify a universal winner.

Figure 1 illustrates smoothness choices from a single fixed scenario, chosen by seed and mechanism rather than by performance. Figure 2 summarizes the differing boundary behavior of the maps. Figure 3 displays their predictive behavior separately for stationary and changing latent roughness. The common branch representation supports these distinct decisions without redefining the penalized trend estimator or carrying forward any validation-stage fitted trend.

Together these experiments support the *representational* contribution: tracking local minima yields

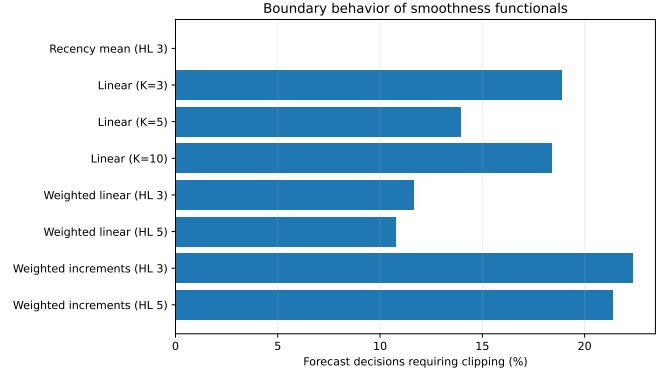


FIGURE 2 Fraction of Checkpoint 08 decisions whose raw smoothness functional requires truncation to $[0, 1]$. Linear and increment extrapolations need not preserve the interval, whereas convex recency averages do.

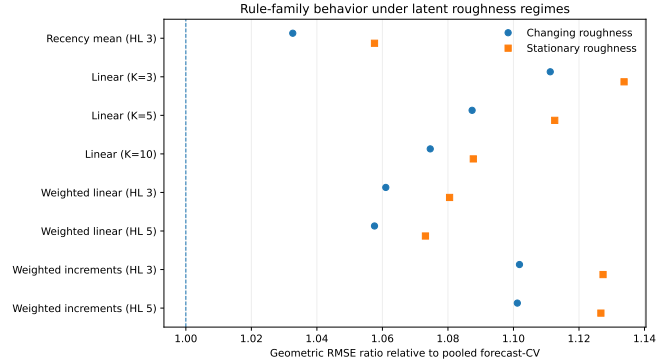


FIGURE 3 Geometric forecast-error ratios against pooled forecast-CV for branch-to-smoothness rules in stationary and changing roughness regimes. Values above one favor the pooled comparator. The plot describes the behavior of a family of rules, not a data-driven winning-rule selection.

a chronologically organized state on which multiple admissible smoothness-selection functionals can be defined and compared under the same information restrictions. They do not support a claim that arbitrary branch-based selection uniformly improves out-of-sample forecasts.

8 | SCOPE, INTERPRETATION, AND IMPLICATIONS

8.1 | What the dynamic smoothness object means

The selected smoothness is not treated as a fixed property of a series. At a given origin, several local minima of the forecast-loss surface may coexist. Their temporal branches summarize recurring smoothing regimes discovered by the data.

For a final selected value $\hat{S}_T$, the effective-degrees-of-freedom interpretation remains

$$\hat{S}_T = \frac{L - \text{edf}_T}{L - d}.$$

A value near zero corresponds to a flexible trend; a value near one corresponds to a trend close to the null-space polynomial. The dynamic contribution is that today's value can depend on the history of a persistent local-minimum branch rather than on one pooled optimum alone.

8.2 | Branches are data-driven, not fixed smoothness bins

The continuation radius ε does not define a preassigned category such as “smoothness around 0.9.” Branch locations arise from local minima of successive forecast-loss surfaces. The radius only determines whether two neighboring minima are close enough to be interpreted as the same branch.

Consequently, a branch can drift over time. Its identity reflects local continuity of the optimum, not membership in a fixed interval.

8.3 | What may be averaged

The framework distinguishes three objects that should not be conflated. Historical forecast losses may be averaged to rank candidate smoothness values or branches. Historical smoothness values may be averaged when a declared branch functional such as a recent-mean rule is used. Historical fitted trends are never averaged in the proposed procedure.

After the current smoothness is selected, the trend is always refit on the newest information available before the forecast.

8.4 | Static and dynamic targets

The pooled selector asks which single value of S performs best on average over historical forecast origins. The branch formulation asks how a time-indexed collection of local minima can be organized and converted into multiple admissible current values of S . Predictive improvement is a possible outcome of particular decisions, not a property guaranteed by the representation itself.

Neither target is automatically the smoothness that best reconstructs a latent historical trend. Forecasting and signal recovery remain distinct objectives.

8.5 | Numerical boundary

The forecasting paper does not claim that a particular root-search or assignment algorithm is the substantive contribution. Recovery of multiple minima, exact endpoint handling, and temporal correspondence of minima belong to the companion numerical-methods paper.

8.6 | Boundary behavior of smoothness forecasts

Normalized smoothness is restricted to $[0, 1]$. Convex means of historically admissible values remain in that interval, whereas linear or increment extrapolations of the branch trajectory need not. Truncation back to the admissible interval is therefore part of defining these functionals, not an innocuous plotting convention. The simulations show that the frequency of truncation varies substantially with the form of ϕ . This is a concrete difference between equally admissible constructions of the next smoothness value.

8.7 | Limits of predictive interpretation

A favorable four-series confirmation result is not sufficient to establish universal superiority. The broad financial panel and controlled changing-roughness experiments did not reproduce an aggregate advantage over pooled forecast-CV. Their errors are evaluated on different scales, and the series within the financial panel are not independent. Accordingly, the paper claims a reusable representation for chronologically tracked

forecast-loss minima and an illustrative family of decision maps; it does not claim that tracked branches inevitably improve forecasting.

8.8 | Limitations

The baseline estimator assumes equally spaced observations, a fixed difference order, a fixed final estimation window, and deterministic continuation of the estimated trend. Checkpoints 05 and 06 show that high-order native polynomial continuation can amplify small differences in selected S into extreme forecast errors. In particular, restricting higher difference orders reduced the catastrophic error tail in a post-hoc diagnostic, but does not validate a revised forecasting selector. The current branch construction also introduces design quantities such as the tracking radius, recent-history length, and weighting parameters. Those quantities must be selected on development data and frozen before outer-test evaluation.

Local-minimum identity can become ambiguous when branches approach or cross. The initial greedy one-to-one matcher is therefore a transparent baseline rather than a mathematical definition of true branch identity. Sensitivity to the numerical correspondence rule must be reported separately from forecasting performance.

Finally, the framework isolates trend extrapolation. It does not yet add a separate stochastic residual forecast model. That restriction keeps the interpretation of any forecast gain tied to smoothness selection rather than to another predictive component.

9 | CONCLUSION

In penalized trend estimation, smoothness is usually represented by a single regularization parameter. The normalized smoothness coordinate offers an interpretable alternative, but chronological forecast-loss surfaces may admit several local minima whose locations vary across forecast origins. Treating these minima as unrelated tuning outcomes discards their temporal structure.

This paper formulates that structure through tracked branches. For each branch, a history of smoothness values and successive Validation-1 and Validation-2 forecast losses defines a state V_j . The decomposition

$$\hat{j}_T = \psi(V_1, \dots, V_J), \quad \hat{S}_T = \phi(V_{j_T})$$

separates selecting a historically credible branch from deciding how its information should produce today's smoothness. The map ϕ need not be unique: the newest minimum, convex recency averages, predictive-loss weights, and extrapolations of the smoothness trajectory are all well-defined examples. This family of decisions is the central methodological contribution, not the identification of a universally optimal smoothing rule.

The empirical record sets clear limits on forecasting claims. Horizon-matched pooled forecast-CV is useful in controlled simulation. The frozen recency-based branch rule improves on pooled CV in one small confirmation panel, but does not retain that advantage on a broader financial panel or in the simulation with changing latent roughness. High-order polynomial continuation also introduces a stability problem that is separate from tracking itself. Finally, the trajectory-rule demonstrations reveal meaningful differences in boundary behavior and forecast error, without establishing superiority for the extrapolative rules.

After every smoothness decision, the operational trend is estimated afresh on the latest available data; historical validation fits are never averaged into the forecast. This chronology makes the proposed branch representation usable for future decision rules and empirical settings while preserving the distinction between a flexible forecasting framework and an empirically validated predictive gain. The companion numerical study addresses the separate task of recovering local minima and maintaining their correspondence through time.

DATA AVAILABILITY

The numerical experiments and their fixed simulation seeds are documented in the accompanying repository. The financial evaluation uses a frozen historical market-data snapshot with a machine-readable manifest; data availability and licensing remain subject to the original providers. Results using revised macroeconomic series are exploratory rather than claims of real-time-vintage forecasting performance.

CODE AVAILABILITY

Reusable implementations of finite-difference penalized trend estimation, chronological forecast objectives, rolling-origin validation, and smoothness transformations are maintained in the `trend_estimation` package at <https://github.com/heri-espino/Trend-Estimation>. The repository also contains the manuscript source, experiment protocols, and frozen result artifacts used for the paper.

CONFLICT OF INTEREST

The author declares no conflict of interest.

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